

Deep Learning Optimized on Jean Zay

Profiler

IDRIS

BY NC SA

Benchmarking vs Profiling

Benchmarking vs Profiling

BENCHMARKING = How fast ?

- User metrics on the whole run: time / step, samples / s, tokens / s etc.
- No instrumentation, low impact
- Compare configurations

PROFILING = Why, and where ?

- Time and resources broken down per phase, op, kernel
- Instrumented, degraded performance
- Locate the bottleneck



Key metrics

BENCHMARKING = How fast ?

Throughput	MFU	Time per step	Memory
samples / s: the number that matters	achieved FLOP/s ÷ GPU peak FLOP/s	Seconds per training iteration	Max GPU memory used

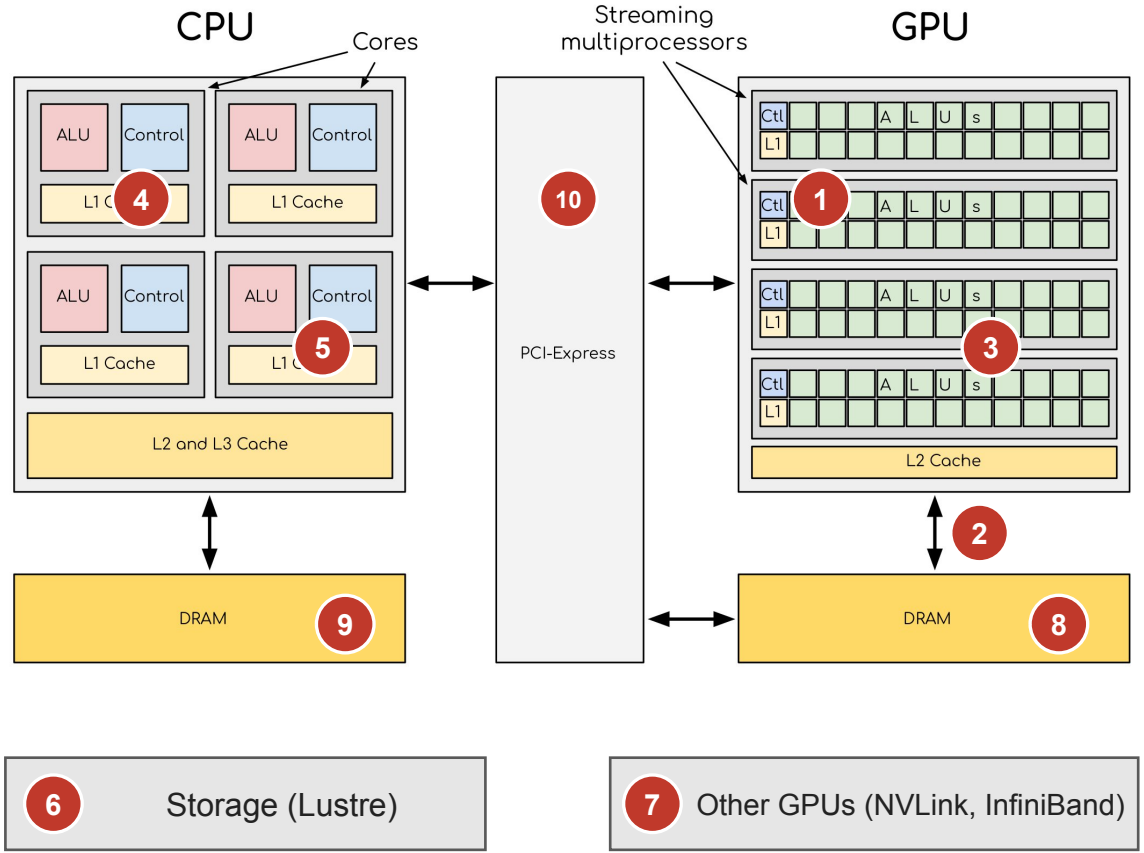
PROFILING = Why, and where ?

SM occupancy Tensor Cores	GPU utilization	Time breakdown	Bandwidth
Are the compute units well used ?	% of time kernel runs	Input wait, compute, communication, per op and kernel etc...	HBM, PCIe , interconnect

Bottlenecks

Where bottlenecks live

https://enccs.github.io/gpu-programming/2-gpu-ecosystem/#id1



Does the GPU compute well ?

1. Compute
2. Memory bandwidth
3. Latency / occupancy

Is the GPU fed ?

4. Kernel launch
5. Data loading
6. Storage I/O
7. Communication

Does the hardware cap it ?

8. GPU memory (VRAM)
9. Host memory (RAM)
10. PCIe transfers

1 · Compute-bound

DEFINITION

More math than the GPU can do.

CAUSE

Big matrix multiplications and convolutions, or FP32 math that skips the Tensor Cores

CONSEQUENCE

Good near peak. On the FP32 path, most of the hardware is wasted

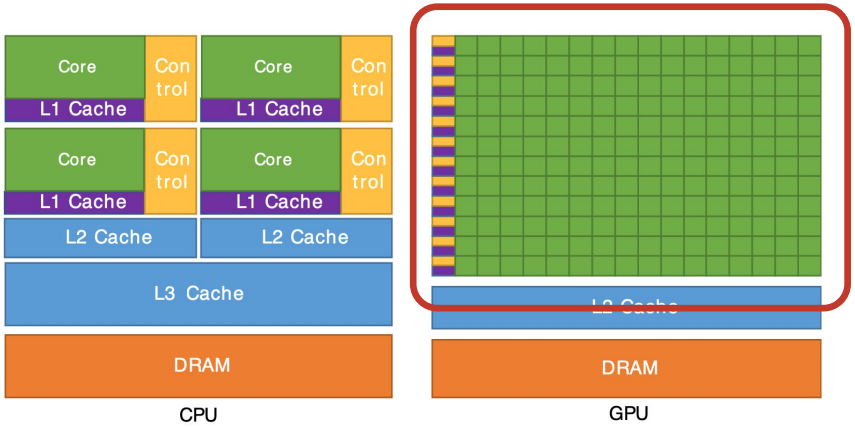
SYMPTOMS

High compute throughput, lower memory throughput.

FIX

Mixed precision (bf16 / fp16), dimensions multiple of 8, fused kernels

https://docs.nvidia.com/cuda/cuda-c-programming-guide/



Thousands of ALUs and Tensor Cores: all busy

2 · Memory-bandwidth-bound

DEFINITION

The GPU waits for data from its own memory (HBM).

CAUSE

Few operations per byte: element-wise ops, normalizations, reductions, casts

CONSEQUENCE

The GPU looks busy, but MFU stays low

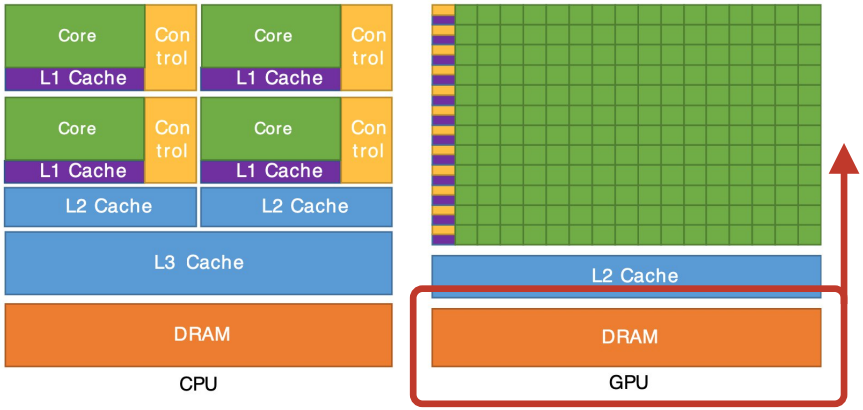
SYMPTOMS

High memory throughput, low compute throughput

FIX

Kernel fusion (torch.compile), smaller data types

https://docs.nvidia.com/cuda/cuda-c-programming-guide/



HBM ≈ 2 TB/s… still too slow to feed every ALU

3 · Latency / occupancy-bound

DEFINITION

Not enough parallel work to keep the GPU busy.

CAUSE

Tiny kernels, small batches, chains of dependent operations

CONSEQUENCE

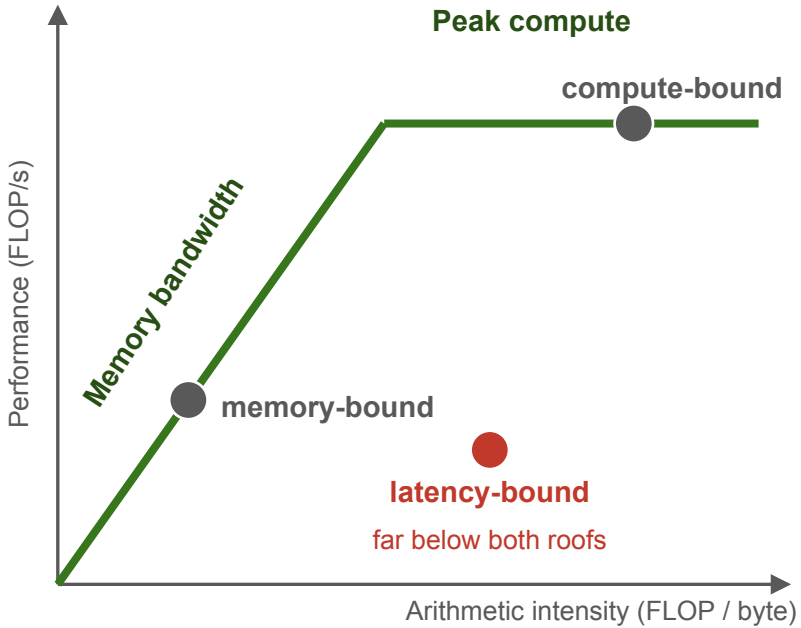
Neither compute nor memory is saturated

SYMPTOMS

Low occupancy, short kernels, both throughputs low

FIX

Larger batch, fuse small kernels



4 · Kernel-launch-bound

DEFINITION

The CPU cannot launch GPU kernels fast enough.

CAUSE

Python overhead, thousands of tiny operations

CONSEQUENCE

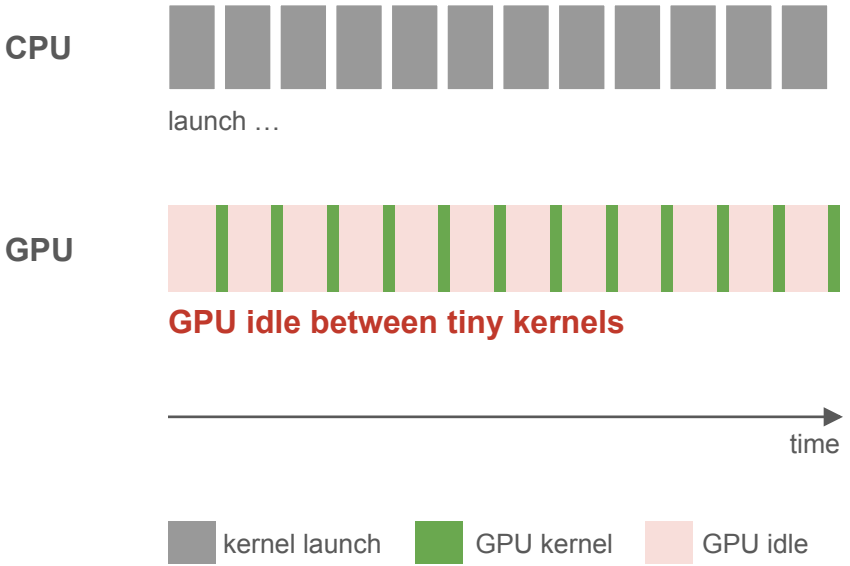
The GPU sits idle between kernels

SYMPTOMS

Gaps between kernels on the GPU timeline, CPU busy launching

FIX

torch.compile, bigger batches



5 · Data-loading-bound

DEFINITION

The input pipeline cannot deliver batches fast enough.

CAUSE

Too few DataLoader workers, heavy CPU decoding and augmentation

CONSEQUENCE

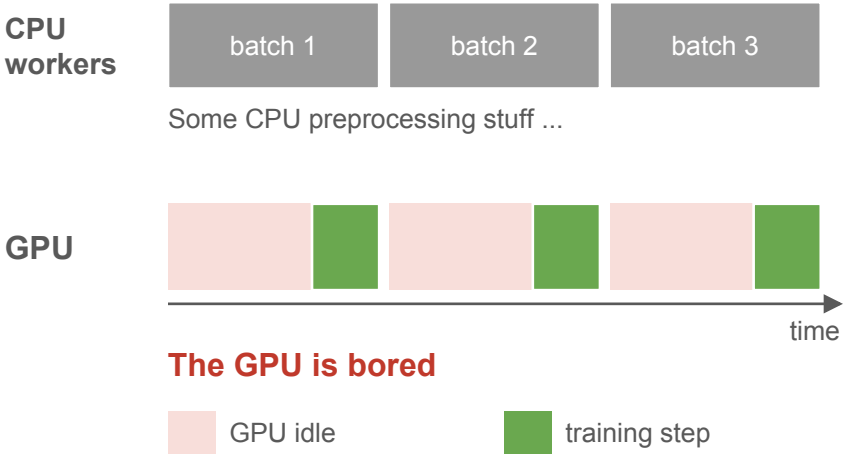
The GPU waits at the start of every step

SYMPTOMS

Long input wait, saw-tooth GPU utilization

FIX

More workers, pin_memory, prefetching, GPU augmentation (DALI)



6 · Storage I/O-bound

DEFINITION

Reading the dataset from disk is the limit.

CAUSE

Millions of small files on a parallel file system (Lustre)

CONSEQUENCE

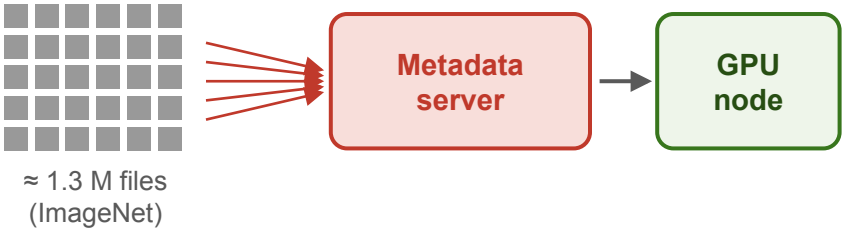
Metadata server saturated: slow first epoch, input stalls

SYMPTOMS

Long open / stat times, disk bandwidth not saturated

FIX

Sharded formats (WebDataset, HDF5): fewer, bigger files



7 · Communication-bound

DEFINITION

GPUs spend their time exchanging data instead of computing.

CAUSE

Gradient all-reduce (DDP, FSDP), slow links between nodes, no overlap

CONSEQUENCE

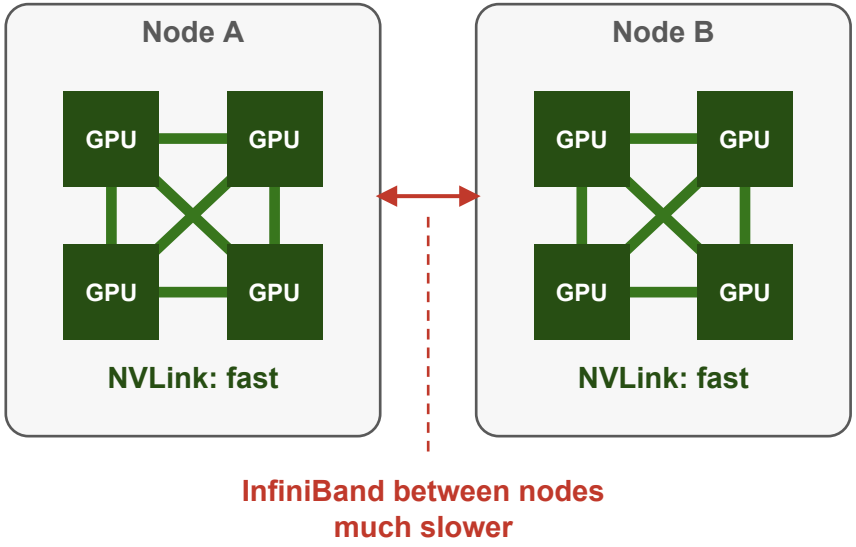
Adding GPUs brings less and less speed-up

SYMPTOMS

Long NCCL kernels, poor compute / comm overlap, sub-linear scaling

FIX

Overlap comm and compute, gradient accumulation, keep traffic on NVLink



8 · GPU memory capacity (VRAM)

DEFINITION

The model and its training state do not fit in GPU memory.

CAUSE

Parameters + gradients + optimizer states + activations > VRAM

CONSEQUENCE

CUDA out-of-memory, or workarounds that slow training down

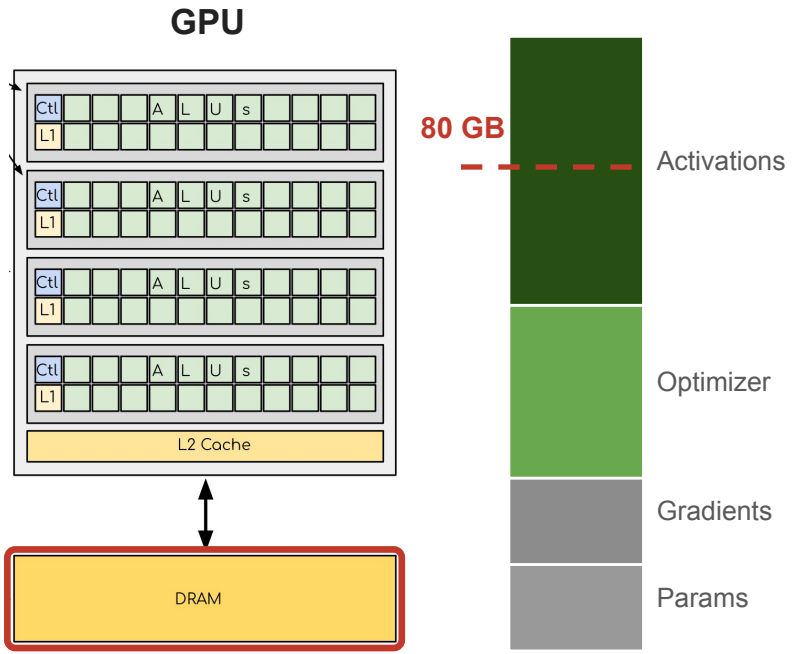
SYMPTOMS

OOM errors, peak memory close to capacity

FIX

AMP, activation checkpointing, FSDP / ZeRO sharding, gradient accumulation

https://enccs.github.io/gpu-programming/2-gpu-ecosystem/#id1



9 · Host memory capacity (RAM)

DEFINITION

The node's CPU memory runs out.

CAUSE

Whole dataset cached in RAM, memory leaks, workers duplicating Python data

CONSEQUENCE

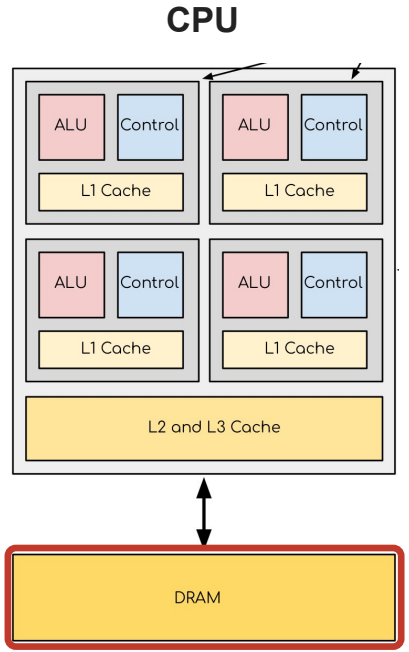
Job killed by Slurm (out of memory), often with no Python error

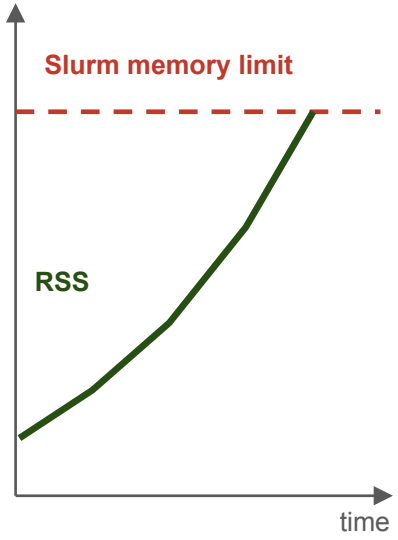
SYMPTOMS

Process memory keeps growing up to the job limit

FIX

Stream the data, fix leaks (.item(), .detach()), NumPy arrays over Python lists





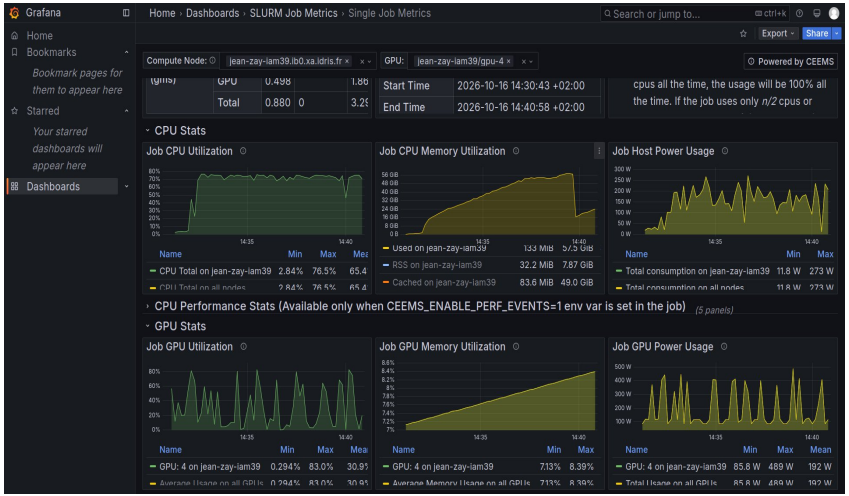
Tools

nvidia-smi & CEEMS

```
+-----------------------------------------+------------------------+----------------------+
|   5  NVIDIA A100-SXM4-80GB          On  |   00000000:8B:00.0 Off |                    0 |
| N/A   44C    P0            158W /  400W |   25087MiB /  81920MiB |     14%      Default |
|                                         |                        |             Disabled |
+-----------------------------------------+------------------------+----------------------+
|   6  NVIDIA A100-SXM4-80GB          On  |   00000000:C8:00.0 Off |                    0 |
| N/A   35C    P0             87W /  400W |    4105MiB /  81920MiB |      1%      Default |
|                                         |                        |             Disabled |
+-----------------------------------------+------------------------+----------------------+
|   7  NVIDIA A100-SXM4-80GB          On  |   00000000:CB:00.0 Off |                    0 |
| N/A   47C    P0            446W /  400W |    6311MiB /  81920MiB |     96%      Default |
|                                         |                        |             Disabled |
+-----------------------------------------+------------------------+----------------------+

+-----------------------------------------------------------------------------------------+
| Processes:                                                                              |
|  GPU   GI   CI              PID   Type   Process name                      GPU Memory |
|        ID   ID                                                             Usage      |
|=========================================================================================|
|    3   N/A  N/A         1274469      C   ...pu-2.8.0+py3.12.11/bin/python      26638MiB |
|    4   N/A  N/A         1283293      C   python                                 5876MiB |
|    5   N/A  N/A         1275078      C   ...94xq/envs/cdl-a100/bin/python      25078MiB |
|    6   N/A  N/A         1286427      C   ...pu-2.8.0+py3.12.11/bin/python       4080MiB |
|    7   N/A  N/A         1278167      C   python                                 6294MiB |
+-----------------------------------------------------------------------------------------+
```

nvidia-smi: live view of the GPU, on the compute node



CEEMS: job history (CPU, GPU, memory, energy)

Best for

- Is the GPU fed ? (4, 5)
- 8 · VRAM
- 9 · Host RAM

TraceML

README | Code of conduct | Contributing | Apache-2.0 license | Security

TraceML

Diagnose slow PyTorch training with zero-code instrumentation. Catch regressions in CI.

Works automatically with: Hugging Face Trainer · PyTorch Lightning · RF-DETR

pypi v0.5.0 | CI passing | python 3.10+ | license Apache 2.0 | stars 188

Quickstart · What you get · Compare runs · Regression checks · Integrations · Documentation

⭐ If TraceML helps you find a bottleneck, please star the repository.

TraceML shows where each training step goes—input loading, data transfer, forward, backward, and optimizer work—then identifies the bottleneck and saves evidence for local comparison or CI.

https://traceopt.ai/

Step-time breakdown and a verdict

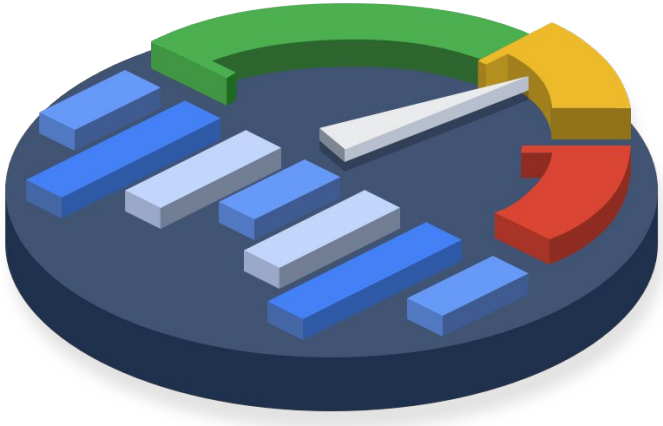
Input wait, H2D, compute, memory per step

Best for

- **5 · Data loading**
- **10 · PCIe transfers**
- **8 · VRAM**
- **6 · Storage I/O (hint)**

PyTorch Profiler

TensorBoard



TensorBoard: operator and kernel views (legacy)

Perfetto: CPU and GPU timeline in the browser

Best for

1 · Compute | 4 · Launch | 5 · Data loading | 8 · VRAM

NVIDIA Nsight Systems & Compute



Nsight Systems

Best for

4 · Launch

7 · Comm



Nsight Compute

Best for

1 · Compute

2 · Memory BW

3 · Latency

TP Time !

